

Algebra II with Trigonometry

Chapter 5.4

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

1. **Principal Value Check:** What is $\sin^{-1}\left(-\frac{1}{2}\right)$ in radians?
2. **Arccos Range:** What is $\cos^{-1}(-1)$ in radians?
3. **Arctan Value:** What is $\tan^{-1}(1)$ in radians?
4. **Defined or Not?:** Is $\sin^{-1}(2)$ defined?
5. **Exact Composition:** What is $\cos\left(\tan^{-1}\left(\frac{3}{4}\right)\right)$?
6. **Exact Composition:** What is $\cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$?
7. **Inverse Relationship:** If $\theta = \cos^{-1}\left(\frac{1}{2}\right)$, what is $\cos \theta$?
8. **Right Triangle Angle:** In a right triangle, the opposite side is 3 and the adjacent side is 4. What inverse trig expression gives the angle?
9. **Shadow Problem:** A tree is 96 ft tall and casts a 120 ft shadow. Which inverse trig function would you use to find the angle of elevation of the sun?
10. **Function Formula:** An observer is 2 miles from a launch pad and sees a shuttle at height h . Write θ as a function of h .